

DHAMOTHARAN N

AI Engineer | Backend Engineer | Full-Stack Developer

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Professional Summary

Software Engineer with 2+ years of experience building AI-powered applications and enterprise software using Python. Experienced in developing scalable backend systems, integrating Generative AI and Large Language Models (LLMs), and building automation solutions from idea to production. Passionate about solving real-world problems through reliable, high-quality software.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL

Backend: Django, FastAPI, Flask, REST APIs, WebSockets, Celery

Frontend: React, TypeScript, HTML5, CSS3, Bootstrap, Tailwind CSS

AI / LLM: OpenAI API, Gemini API, LangChain, RAG, AI Agents, Prompt Engineering, GPT-4 Vision

Computer Vision: OpenCV, TensorFlow, DeepFace, Tesseract OCR

Databases: PostgreSQL, MySQL, SQLite, Redis, ChromaDB, FAISS, Milvus

Cloud & DevOps: AWS, Docker, Nginx, Git, Linux, Cloudflare Workers, Cloudflare D1, Durable Objects

Professional Experience

Software Developer – M7 Corporation, Chennai

Sep 2024 – Present

- Designed production AI platforms using Python, Django, FastAPI, and OpenAI APIs; built multilingual AI assistants and AI-powered government applications.
- Developed scalable REST APIs, webhook integrations, background workers, RAG pipelines, and computer vision/OCR workflows.
- Managed cloud deployments, DevOps automation, and infrastructure optimization while converting business requirements into production products.

Software Engineer – Lmark & Revobig Technology, Chennai

Mar 2024 – Sep 2024

- Built enterprise web applications using Django, JavaScript, and MySQL; designed backend architectures and business automation modules.
- Developed financial automation tools, third-party API integrations, and improved performance through debugging and production support.

Key Projects

Enterprise HR & Workforce Management Platform

Category: Enterprise SaaS — HRMS — AI-powered Attendance System

Multi-tenant HR platform for employee lifecycle, attendance, leave, and payroll with role-based access control. Built AI-powered facial recognition attendance with GPS validation, multi-shift monitoring, employee analytics, and scalable REST APIs.

Stack: Python, Django, PostgreSQL, Redis, Celery, TensorFlow, DeepFace, OpenAI API.

Deployed for: M7 Tech, Thaagam Foundation, GCC.

Thaagam Social Impact Platform

Category: NGO Management — Crowdfunding — Volunteer Platform

End-to-end NGO platform for donor engagement, crowdfunding, volunteer management, and inventory, with AI-powered media processing that reduced generation time from 2–3 minutes to 30–60 seconds.

Stack: Python, Django, PostgreSQL, Redis, Gemini API, OpenCV, Razorpay.

Deployed for: Thaagam Foundation.

AI Civic Communication & Grievance Platform

Category: GovTech — Smart City — AI Grievance Management

Multilingual WhatsApp-based AI assistants and grievance management for municipal corporations, featuring RAG-based knowledge retrieval, AI image validation, and intelligent complaint routing.

Stack: Python, Django, FastAPI, LangChain, OpenAI API, Gemini API, PostgreSQL, Redis.
Deployed for: Tambaram, Hosur, Vellore, Karur, GCC.

AI Vision & Document Intelligence Platform

Category: Computer Vision — OCR — Document AI
Computer vision and document intelligence platform for complaint validation, OCR, and attendance verification, extracting structured data from scanned and handwritten documents.
Stack: Python, OpenCV, TensorFlow, GPT-4 Vision, Tesseract OCR.
Deployed for: GCC, HAC, Impact Foundation.

Social Intelligence & X Analytics Platform

Category: Social Media Analytics — Sentiment Analysis — GovTech
Real-time social media intelligence platform for complaint monitoring, sentiment analysis, viral post detection, and governance analytics dashboards.
Stack: Python, Django, PostgreSQL, Redis, Celery, X API.
Deployed for: GCC, Tambaram Corporation.

CommAI – AI Communication & Hospital Automation Platform

Category: Healthcare AI — Conversational AI — Hospital Automation
AI communication platform integrating WhatsApp, web portals, and LLM assistants for hospital appointment booking, patient communication, automated reminders, and workflow automation.
Stack: Python, Django, FastAPI, OpenAI API, Gemini API, WhatsApp Business API.
Deployed for: Bharath Hospital, Thaagam Foundation.

Personal Projects

DriveManager – Multi-Account Google Workspace Platform

Centralized platform for managing multiple Google Workspace accounts, file transfers, and authentication, integrating OAuth, Drive, Gmail, Docs, Sheets, and Photos with secure token management.
Stack: Python, Flask, React, TypeScript, SQLite, Google Drive API, Docker

OpenTrack Manager – Project Monitoring Platform

Project tracking and deployment management platform with Python backend services, deployment automation, and Docker-based workflows for monitoring project status.
Stack: Python, TypeScript, Docker, HTML, CSS

OpenDevOps – Cloud Deployment Platform

Cloud-native DevOps platform for infrastructure automation, deployment orchestration, and project management with reusable, scalable TypeScript services.
Stack: TypeScript, Python, HTML

Education

B.Sc. Computer Science (Distance Education)	<i>2025 – Present</i>
Alagappa University, Tamil Nadu	
B.A. LL.B	<i>2017 – 2021</i>
North East Frontier Technical University, Arunachal Pradesh	

Certifications

Python Full Stack Development	<i>2022 – 2023</i>
Greens Technology, Chennai	